

A random-effects hurdle model for predicting bycatch of endangered marine species

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GSEM, October 9th, 2018

Motivation

Some facts:

- For 1kg of plaice or sole to be commercialized, there is about 6kg of bycatch. This amount varies between 10 and 40 kg for shrimps.
- According to WWF, every year 300'000 whales and daulphins, 100 millions of sharks and 300'000 sea birds are accidentally caught.
- Underperformance of current fishing practice (with respect to a sustainable alternative) costs 50 million dollars annually (World Bank and FAO).

Motivation

Convention on International Trade in Endangered Species of Wild Fauna and Flora (CITES) press release (Bangkok, 14 March 2013):

“CITES conference takes decisive action to halt decline of tropical timber, sharks, manta rays and a wide range of other plants and animals.
[...]

The Parties confirmed a decision made by one of the Conferences Committees earlier in the week to include five commercially valuable shark species in Appendix II. The oceanic whitetip (*Carcharhinus longimanus*), scalloped hammerhead (*Sphyrna lewini*), great hammerhead shark (*Sphyrna mokarran*), smooth hammerhead shark (*Sphyrna zigaena*) and the porbeagle shark (*Lamna nasus*) are harvested in huge numbers for their valuable fins and, in some cases, meat. From now onwards, they will have to be traded with CITES permits and evidence will have to be provided that they are harvested sustainably and legally. These listings mark a milestone in the involvement of CITES in marine species. ”

Motivation



October is National Seafood Month



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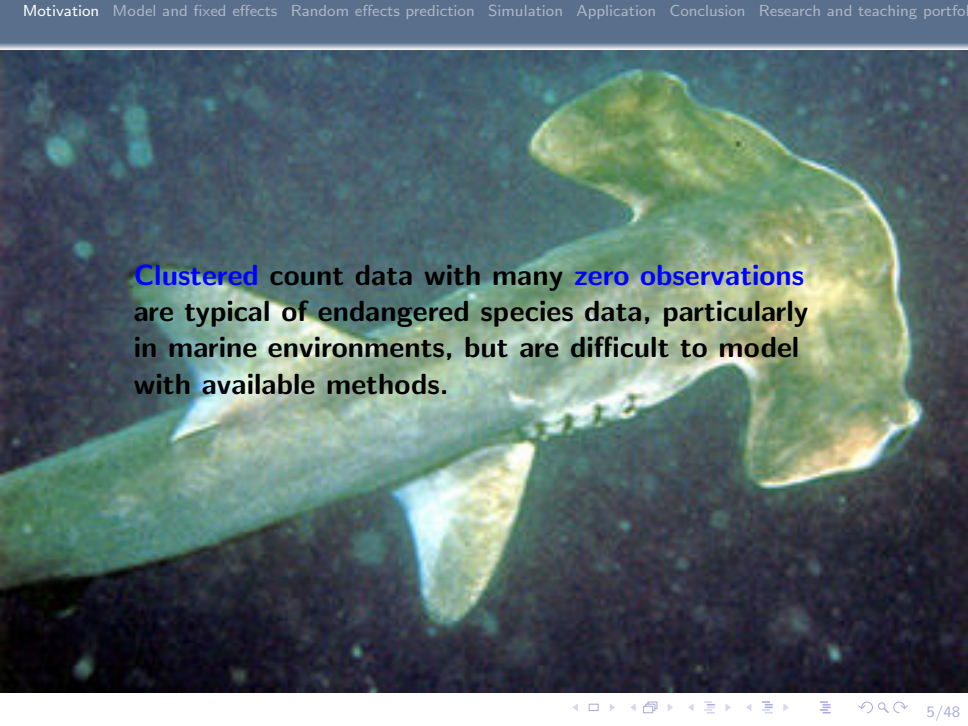
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NEWS

2018 Bycatch Reduction Engineering Program Awards

August 15, 2018

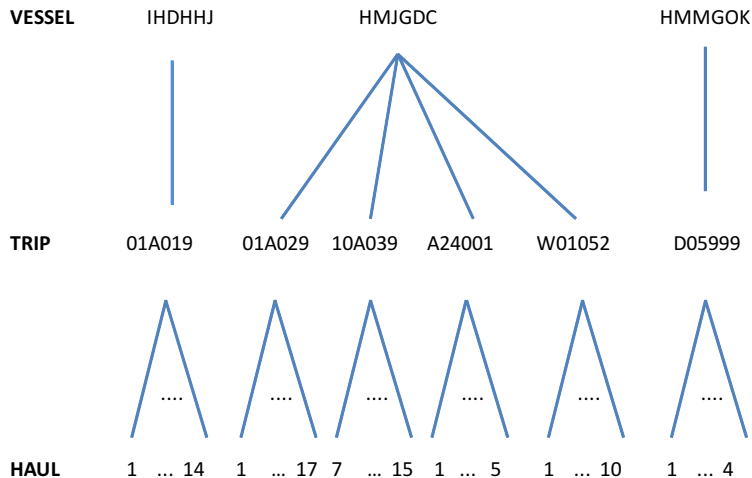
More than \$2.3 million will go to partners to support innovative bycatch reduction research projects.

A hammerhead shark is shown swimming in dark, deep water. The shark's head is wide and flat, with a prominent white patch on its snout. Its body is greyish-brown, and its fins are visible. The background is dark and slightly out of focus, showing some light-colored particles or bubbles.

Clustered count data with many **zero observations** are typical of endangered species data, particularly in marine environments, but are difficult to model with available methods.

Motivation

Clustering:



Motivation

Counts with excess zeros:

value	frequency
0	1542
1	145
2	53
3	20
4	16
5	10
6	4
7	8
8	3
9	3
10	2
11-20	13
20-50	6

Mission

Our goals were:

- Model the data (taking into account the excess of zeros and the clustering simultaneously)
- Fit the model
- (Optimally) predict at the observation level (empirical best predictors for cluster specific targets)
- Provide MSEP of the predictions

Outline

- 1 Motivation
- 2 Model and fixed effects
- 3 Random effects prediction
- 4 Simulation
- 5 Application
- 6 Conclusion
- 7 Research and teaching portfolio

Hurdle model for independent data

Independent (count) responses y_i for $i = 1, \dots, n$. Set of covariates $\mathbf{x}_i$ and $\mathbf{z}_i$, possibly overlapping. Then:

$$P(Y_i = y_i) = [y_i] = \begin{cases} 1 - p(\mathbf{x}_i) & y_i = 0 \\ p(\mathbf{x}_i)f(y_i, \lambda(\mathbf{z}_i)) & y_i = 1, 2, 3, \dots \end{cases}$$

where $f(y_i, \lambda(\mathbf{z}_i))$ is a discrete distribution on the positive integers.

Model $p(\mathbf{x}_i)$ and $\lambda(\mathbf{z}_i)$ as

$$\text{logit}\{p(\mathbf{x}_i)\} = \mathbf{x}_i^T \boldsymbol{\alpha}$$

$$\log\{\lambda(\mathbf{z}_i)\} = \mathbf{z}_i^T \boldsymbol{\beta}.$$

The two parts of the model can be fitted separately by maximum likelihood.

Random effects models

Responses y_{ij} for $i = 1, \dots, c$ clusters, and $j = 1, \dots, n_i$ observations per cluster. Independence between clusters.

1. Linear mixed-model:

$$y_{ij} = \mathbf{z}_{ij}^T \boldsymbol{\beta} + \mathbf{w}_{ij}^T \mathbf{u}_i + \epsilon_{ij}.$$

Fit by maximum likelihood (or REML). Optimal predictions by BP, BLUE, BLUP.

2. Generalized linear mixed models:

$$\log(E(y_{ij})) = \mathbf{z}_{ij}^T \boldsymbol{\beta} + \mathbf{w}_{ij}^T \mathbf{u}_i.$$

Fit by maximum likelihood (approximations needed).
Predictions by conditional modes. Properties?

Our proposal: hurdle model with random effects

Responses y_{ij} for $i = 1, \dots, c$ clusters, and $j = 1, \dots, n_i$ observations per cluster. Independence between clusters.

Two sets of associated covariates $\mathbf{x}_{ij}$ and $\mathbf{z}_{ij}$, possibly overlapping.

Random effects vectors $\mathbf{u}_i$ ($d_i \times 1$) and $\mathbf{v}_i$ ($e_i \times 1$).

Conditional density of y_{ij} :

$$[y_{ij} \mid \mathbf{u}_i, \mathbf{v}_i] = \begin{cases} 1 - p(\mathbf{x}_{ij}, \mathbf{u}_i) & y_{ij} = 0 \\ p(\mathbf{x}_{ij}, \mathbf{u}_i) f(y_{ij}, \lambda(\mathbf{z}_{ij}, \mathbf{u}_i, \mathbf{v}_i)) & y_{ij} = 1, 2, 3, \dots \end{cases}$$

where $f(y_{ij}, \lambda(\mathbf{z}_{ij}, \mathbf{u}_i, \mathbf{v}_i))$ is a discrete distribution on the positive integers.

Hurdle model with random effects

Model $p(\mathbf{x}_{ij}, \mathbf{u}_i)$ as

$$\text{logit}\{p(\mathbf{x}_{ij}, \mathbf{u}_i)\} = \mathbf{x}_{ij}^T \boldsymbol{\alpha} + \text{random structure}(\mathbf{u}_i)$$

and $\lambda(\mathbf{z}_{ij}, \mathbf{u}_i, \mathbf{v}_i)$ as

$$\log\{\lambda(\mathbf{z}_{ij}, \mathbf{u}_i, \mathbf{v}_i)\} = \mathbf{z}_{ij}^T \boldsymbol{\beta} + \text{random structure}(\mathbf{u}_i, \mathbf{v}_i).$$

General structure for the random effects

Presence/absence model:

$$\text{logit}\{p(\mathbf{x}_{ij}, \mathbf{u}_i)\} = \mathbf{x}_{ij}^T \boldsymbol{\alpha} + \sum_{k=1}^a \sigma_{uk} \mathbf{w}_{ijk}^T \mathbf{u}_{ik}$$

Abundance model:

$$\log\{\lambda(\mathbf{z}_{ij}, \mathbf{u}_i, \mathbf{v}_i)\} = \mathbf{z}_{ij}^T \boldsymbol{\beta} + \sum_{k=1}^{b_r} \mathbf{r}_{ijk}^T \{\gamma_k \sigma_{uk} \mathbf{u}_{ik} + \sigma_{rk} \mathbf{v}_{rik}\} + \sum_{k=1}^{b_s} \sigma_{sk} \mathbf{s}_{ijk}^T \mathbf{v}_{sik}.$$

We next assume

- $\mathbf{u}_i \sim g_i(\mathbf{u}_i, \boldsymbol{\tau}) = \prod_{k=1}^a g_{ik}(\mathbf{u}_{ik}, \boldsymbol{\tau}_k) = \prod_{k=1}^a \prod_{l=1}^{d_{ik}} g_{ikl}(\mathbf{u}_{ikl}, \boldsymbol{\tau}_k).$
- $\mathbf{v}_i \sim h_i(\mathbf{v}_i, \boldsymbol{\kappa}) = \prod_{k=1}^{b_r} h_{rik}(\mathbf{v}_{rik}, \boldsymbol{\kappa}_k) \prod_{m=1}^{b_s} h_{sim}(\mathbf{v}_{sim}, \boldsymbol{\kappa}_m) = \prod_{k=1}^{b_r} \prod_{l=1}^{e_{rik}} h_{rikl}(\mathbf{v}_{rikl}, \boldsymbol{\kappa}_k) \prod_{m=1}^{b_s} \prod_{l=1}^{e_{vik}} h_{siml}(\mathbf{v}_{siml}, \boldsymbol{\kappa}_m)$

Some important special cases

- Random intercept model:

$$\sigma_{u1} u_{i1} , \gamma_1 \sigma_{u1} u_{i1} + \sigma_{v1} v_{i1}$$

- Random slopes and intercept model:

$$\sum_{k=1}^a \sigma_{uk} x_{ijk} u_{ik} , \sum_{k=1}^a r_{ijk} \{ \gamma_k \sigma_{uk} u_{ik} + \sigma_{vk} v_{rik} \} + \sum_{k=1}^{b_s} \sigma_{tk} s_{ijk} v_{sik}$$

- Two-level nested model (multilevel model):

$$\sigma_{u1} u_{i1} + \sigma_{u2} \mathbf{w}_{ij2}^T \mathbf{u}_{i2} , \gamma_1 \sigma_{u1} u_{i1} + \sigma_{v1} v_{i1} + \mathbf{w}_{ij2}^T (\gamma_2 \sigma_{u2} \mathbf{u}_{i2} + \sigma_{v2} \mathbf{v}_{i2})$$

The marginal likelihood

Let θ be the vector of all the parameters α , $\sigma_{u1}, \dots, \sigma_{ua}$, β , γ , $\sigma_{r1}, \dots, \sigma_{rb_r}$, $\sigma_{s1}, \dots, \sigma_{sb_s}$, ν , τ and κ .

The marginal likelihood writes

$$\begin{aligned} L(\theta) = \prod_{i=1}^c [\mathbf{y}_i] &= \prod_{i=1}^c \int \int [\mathbf{y}_i | \mathbf{u}_i, \mathbf{v}_i] g_i(\mathbf{u}_i, \tau) h_i(\mathbf{v}_i, \kappa) d\mathbf{u}_i d\mathbf{v}_i. \\ &= \prod_{i=1}^c \int \int \prod_{j=1}^{n_i} [y_{ij} | \mathbf{u}_i, \mathbf{v}_i] g_i(\mathbf{u}_i, \tau) h_i(\mathbf{v}_i, \kappa) d\mathbf{u}_i d\mathbf{v}_i. \end{aligned}$$

Implementation

Current implementation:

- Likelihood computed numerically by simple Monte Carlo.
- Numerical optimization of the likelihood with `optim` in R. Numerical Hessian matrix also obtained from this function.

Asymptotics of the maximum likelihood estimator

Theorem

Under mild conditions, there is a solution $\hat{\boldsymbol{\theta}}$ of the estimating equations satisfying $|\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0| = O_p(c^{-1/2})$ as $c \rightarrow \infty$. Moreover,

$$c^{1/2}(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0) = c^{-1/2} \mathbf{J}(\boldsymbol{\theta}_0)^{-1} \sum_{i=1}^c \psi_i(\mathbf{y}_i, \mathbf{x}_i, \mathbf{z}_i, \boldsymbol{\theta}_0) + o_p(1),$$

and

$$c^{1/2}(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0) \xrightarrow{D} N(\mathbf{0}, \mathbf{J}(\boldsymbol{\theta}_0)^{-1}).$$

Targets to predict

Beside the random quantities $\mathbf{u}_i$ and $\mathbf{v}_i$ themselves, interesting cluster specific quantities:

- probability of presence given $\mathbf{u}_i$:

$$P(Y_{ij} > 0 | \mathbf{u}_i) = p(\mathbf{x}_{ij}, \mathbf{u}_i),$$

- expected abundance given presence and given $\mathbf{u}_i$ and $\mathbf{v}_i$:

$$E(Y_{ij} | Y_{ij} > 0, \mathbf{u}_i, \mathbf{v}_i),$$

- expected abundance given $\mathbf{u}_i$ and $\mathbf{v}_i$:

$$E(Y_{ij} | \mathbf{u}_i, \mathbf{v}_i) = p(\mathbf{x}_{ij}, \mathbf{u}_i) E(Y_{ij} | Y_{ij} > 0, \mathbf{u}_i, \mathbf{v}_i).$$

Mean squared error

The targets are all of the form $t(\mathbf{u}_i, \mathbf{v}_i, \mathbf{x}, \mathbf{z}, \theta)$.

The mean squared error of prediction writes:

$$\begin{aligned} \text{mse}_{p_i}(\hat{T}_t, t) &= E_{\mathbf{u}_i, \mathbf{v}_i, \mathbf{y}_i} \left(\hat{T}_t(\mathbf{x}_{ij}, \mathbf{z}_{ij}, \hat{\theta}; \mathbf{y}_i) - t(\mathbf{u}_i, \mathbf{v}_i, \mathbf{x}_{ij}, \mathbf{z}_{ij}, \theta) \right)^2 \\ &= E_{\mathbf{u}_i, \mathbf{v}_i, \mathbf{y}_i} \left(T_t(\mathbf{x}_{ij}, \mathbf{z}_{ij}, \hat{\theta}; \mathbf{y}_i) - T_t(\mathbf{x}_{ij}, \mathbf{z}_{ij}, \theta; \mathbf{y}_i) \right)^2 \\ &\quad + E_{\mathbf{u}_i, \mathbf{v}_i, \mathbf{y}_i} \left(T_t(\mathbf{x}_{ij}, \mathbf{z}_{ij}, \theta; \mathbf{y}_i) - t(\mathbf{u}_i, \mathbf{v}_i, \mathbf{x}_{ij}, \mathbf{z}_{ij}, \theta) \right)^2 \end{aligned}$$

Optimal best predictors

The best predictors (minimizing mse) are given by

$$\begin{aligned} T_t(\mathbf{x}, \mathbf{z}, \boldsymbol{\theta}; \mathbf{y}_i) &= \int \int t(\mathbf{u}, \mathbf{v}, \mathbf{x}, \mathbf{z}, \boldsymbol{\theta}) [\mathbf{u}, \mathbf{v} | \mathbf{y}_i] d\mathbf{u} d\mathbf{v} \\ &= \frac{\int \int t(\mathbf{u}, \mathbf{v}, \mathbf{x}, \mathbf{z}, \boldsymbol{\theta}) [\mathbf{y}_i | \mathbf{u}, \mathbf{v}] g(\mathbf{u}, \boldsymbol{\tau}) h(\mathbf{v}, \boldsymbol{\kappa}) d\mathbf{u} d\mathbf{v}}{\int \int [\mathbf{y}_i | \mathbf{u}, \mathbf{v}] g(\mathbf{u}, \boldsymbol{\tau}) h(\mathbf{v}, \boldsymbol{\kappa}) d\mathbf{u} d\mathbf{v}} \end{aligned}$$

We use simple Monte Carlo to obtain empirical best predictors $\hat{T}_t(\mathbf{x}, \mathbf{z}, \hat{\boldsymbol{\theta}}; \mathbf{y}_i)$ ($\boldsymbol{\theta}$ is replaced by $\hat{\boldsymbol{\theta}}$).

Parametric bootstrap

Estimate $\text{mse}_p(\hat{T}_t, t)$ with the following algorithm:

- Compute the estimate $\hat{\theta}$ from the data.
- For $b = 1, \dots, B$
 - use the bootstrap to generate $\hat{\theta}_b^*$ (fast bootstrap, in fact)
 - generate $\mathbf{u}_{b1}^*, \mathbf{u}_{b0}^*$ from $g(\mathbf{u}, \hat{\tau})$, $\mathbf{v}_{b1}^*, \mathbf{v}_{b0}^*$ from $h(\mathbf{v}, \hat{\kappa})$ and y_{bi}^* from the density $[y_i | \mathbf{u}_{b1}^*, \mathbf{v}_{b1}^*]$.
- Compute the bootstrap estimate of $\text{mse}_p(\hat{T}_t, t)$ as

$$\frac{1}{B} \sum_{b=1}^B \left[\hat{T}_t(\mathbf{x}_i, \mathbf{z}_i, \hat{\theta}_b^*, y_{bi}^*) - t(\mathbf{u}_{b0}^*, \mathbf{v}_{b0}^*, \mathbf{x}_i, \mathbf{z}_i, \hat{\theta}) \right]^2.$$

Fast bootstrap details

For an estimator defined by a fixed point equation $\hat{\theta} = \mathbf{g}(\hat{\theta})$, the fast bootstrap estimator is the adjusted one step estimator

$$\hat{\theta}_A^* = \hat{\theta} + \{\mathbf{I} - \partial_{\theta}\mathbf{g}(\hat{\theta})\}^{-1}\{\hat{\theta}^{(1)*} - \hat{\theta}\},$$

where $\hat{\theta}^{(1)*} = \mathbf{g}^*(\hat{\theta})$.

For our model, first order conditions: $\mathbf{0} = \psi(\hat{\theta})$.

Consider $\mathbf{g}(\theta) = \theta + \mathbf{A}(\theta)\psi(\theta)$ with $\mathbf{A}(\theta) = \mathbf{I}$, so that $\partial_{\theta}\mathbf{g}(\theta) = \mathbf{I} + \partial_{\theta}\psi(\theta)$, and hence $\mathbf{I} - \partial_{\theta}\mathbf{g}(\theta) = -\partial_{\theta}\psi(\theta)$.

The fast bootstrap estimator is then

$$\hat{\theta}_A^* = \hat{\theta} + \{-\partial_{\theta}\psi(\hat{\theta})\}^{-1}\{\hat{\theta}^{(1)*} - \hat{\theta}\},$$

with $\hat{\theta}^{(1)*} = \hat{\theta} + \psi^*(\hat{\theta})$.

Asymptotics of the fast bootstrap estimator

Theorem

Under quite general conditions, we have that

$$c^{1/2}(\hat{\boldsymbol{\theta}}_A^* - \hat{\boldsymbol{\theta}}) \rightarrow N(\mathbf{0}, \mathbf{J}(\boldsymbol{\theta}_0)^{-1} \mathbf{F} \mathbf{J}(\boldsymbol{\theta}_0)^{-1}).$$

The asymptotic distribution of the fast bootstrap estimator is the same as that of the maximum likelihood estimator, when the model is correct and regular (so we can interchange the order of integration and differentiation).

Simulation

Evaluate our **dependent** hurdle mixed effects model (with correlated random effects) and contrast it to the **independent** hurdle model which fits the two parts of the hurdle model separately.

truncated Poisson hurdle model with random intercepts
 $u_{i1} \sim N(0, 1)$ and $v_{i1} \sim N(0, 1)$.

$c = 100$ clusters (half with 5 and half with 10 measurements).

$\mathbf{x}_{ij} = (1, N(0, 1), \text{Bernoulli}(1/2))$ and

$\mathbf{z}_{ij} = (1, N(0, 1), \text{Bernoulli}(1/2), N(0, 1))$.

400 simulations, $K = 1000$ (likelihood approximation)

$B = 1000$ bootstraps.

Four parameters settings (proportion of zeros of either 0.3 or 0.75 on average, dependent or independent hurdle generating model).

Simulation messages

Parameter estimation:

- For the **dependent** hurdle model:
 - All the regression parameters are estimated unbiasedly in the four settings. The dependence parameter is also approximately unbiased, but with large variability. The “variance components” are slightly underestimated.
 - The confidence intervals for the regression parameters meet the nominal coverage, even more so for the independent setting. The results for the parameters of the random components are less satisfactory.
- For the **independent** hurdle model:
 - some of the regression parameters are biased when the data generating process is dependent (in particular, in the abundance part of the model).
 - Due to software restriction (SE availability), confidence interval are available only for the regression parameters. They perform ok.

Simulation messages

Prediction, for the **dependent** hurdle model:

- Small bias, more often negative, due to the underestimation of the spread parameters.
- Agreement between average estimated msep and empirical msep through simulation (with some exceptions).
- Normal prediction intervals for random intercepts show good coverage for u_i (presence/absence equation), but not so good for v_i (abundance equation).

In summary: ignoring the dependence can lead to bias in the fixed-effects regression parameters as well as an inability to predict random effects in some cases.

Hammerhead sharks dataset

1825 observations on 292 different trips (1 to 21 hauls per trip).

Bycatch counts: 85% zeros, positive counts from 1 to 46.

Covariates:

- the year (YEAR, from 1 to 14, representing the period 1992-2005)
- average hook depth (AVGHKDEP, from 6.40 to 182.88)
- area (4=South Atlantic Bight and 5=Mid Atlantic Bight)
- season (SEASON, 464 observations in autumn, 543 in spring, 525 in summer and 293 in winter)
- catch effort: the logarithm of the number of hooks (TOTHOOK, from 25 to 1548).

Model for hammerhead sharks

Truncated Poisson hurdle model with random intercepts

$u_{i1} \sim N(0, 1)$ and $v_{i1} \sim N(0, 1)$.

$K = 5000$ variates for the likelihood approximation.

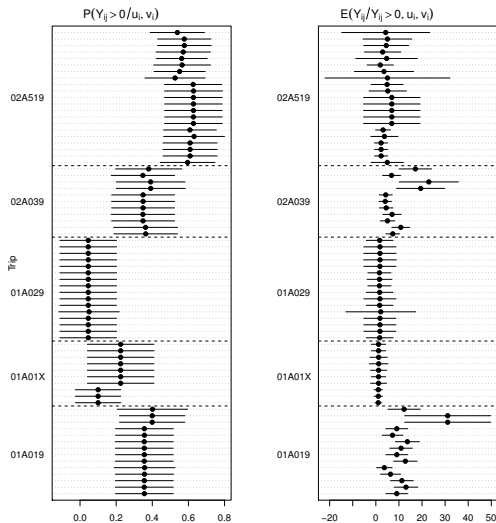
30 starting points for the likelihood maximisation (package `fields` in R).

$B = 1000$ bootstrap samples.

Parameter estimates and standard errors

Variable	Presence-absence dep. hurdle model	indep. hurdle model
	Coeff. (SE-Hess, SE-boot)	Coeff. (SE)
Intercept	-2.123 (1.453,1.636)	1.951 (1.508)
YEAR	-0.059 (0.028,0.027)	-0.044 (0.030)
AVGHKDEP	-0.011 (0.011,0.016)	0.007 (0.010)
AREA5	-0.241 (0.242,0.317)	-0.053 (0.254)
SEASONspring	1.609 (0.341,0.351)	1.630 (0.358)
SEASONsummer	0.074 (0.362,0.372)	0.096 (0.366)
SEASONwinter	1.068 (0.369,0.342)	0.950 (0.393)
log(TOTHOOK)	-0.008 (0.212,0.206)	-0.170 (0.222)
Abundance		
Intercept	-4.871 (0.661,0.678)	-3.322 (1.148)
YEAR	-0.132 (0.032,0.029)	-0.105 (0.042)
AVGHKDEP	-0.067 (0.008,0.013)	-0.052 (0.013)
AREA5	-0.151 (0.230,0.257)	-0.182 (0.249)
SEASONspring	0.850 (0.362,0.236)	-0.121 (0.519)
SEASONsummer	-0.435 (0.488,0.401)	-0.843 (0.590)
SEASONwinter	1.278 (0.342,0.226)	0.288 (0.574)
log(TOTHOOK)	1.020 (0.096,0.122)	0.944 (0.171)
γ_1	1.116 (0.159,0.157)	—
σ_{u1}	1.413 (0.145,0.134)	1.387 (n.a.)
σ_{v1}	1.248 (0.144,0.179)	1.544 (n.a.)

Prediction of probability of presence and conditional abundance



Final thoughts

For the random-effects hurdle model:

- Implementation extension to other models, in particular other random effects structures.
- Other approximation techniques: for example ADMB or TMB (Laplace approximation by automatic differentiation).
- Other distributions for random effects.
- Applies to other fields: health economics, medicine, educational science.

More generally:

- General procedure (optimal predictors and MSEP computation), can be used for example for GLMM.

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